Model	Concept	Spectral Signature	Key References
Pure Photospheric Fireball	GRB fireball becomes transparent at the photosphere; photons escape as thermal radiation.	Nearly Planckian blackbody peak (too narrow compared to observed GRBs).	Goodman (1986); Paczyński (1986)
Multi-Color / Multi-Temperature BB (mBB)	Jet angle/geometry produces temperature variation across the emitting surface.	Superposition of BBs → broadened spectrum.	Ryde (2004, 2005); Pe'er & Ryde (2011)
Sub-Photospheric Dissipation	Dissipation (shocks, magnetic reconnection, collisional heating) below the photosphere modifies the emerging BB.	Comptonized BB: thermal peak + broader high-energy tail.	Rees & Mészáros (2005); Pe'er et al. (2006)
Comptonized Blackbody (BB + PL/CPL)	Observed GRB fits often need a thermal peak plus a non-thermal power-law tail.	BB core with an additional PL or cutoff PL.	Ryde (2004, 2009); Guiriec et al. (2011, 2013)
Two-Component Jet (Spine–Sheath)	Jet has inner spine (thermal photosphere) + outer sheath (shocks, non-thermal).	BB + Band-like non-thermal component.	Lundman (2012)
Hybrid BB + Synchrotron	Photosphere contributes a BB, while accelerated electrons in the jet produce synchrotron radiation.	Thermal peak + broader synchrotron-dominated spectrum.	Burgess et al. (2014); Yu et al. (2019)

A **photospheric model** of gamma-ray bursts describes the emission of radiation from the point in the relativistic outflow where the plasma becomes transparent, known as the photosphere. At the very beginning of the burst, when the outflow has just been launched, the plasma is extremely dense. In this early stage of the fireball's expansion, photons cannot escape freely but are trapped by repeated scattering on electrons and positrons. As the fireball expands and the density decreases, the outflow eventually becomes transparent, allowing photons to escape from this "last scattering surface." The photospheric model therefore explains the presence of thermal or quasi-thermal components in GRB spectra as the natural release of radiation from the photosphere.

Summary of papers by Goodman, 1986; Paczyński 1986

Pure Photospheric Fireball Models

When gamma–ray bursts (GRBs) were first observed, their extreme properties immediately posed a challenge. The bursts are extraordinarily bright, with energies on the order of 10^51 erg released in less than a second. Their spectra peak in the hundreds of keV to MeV range, and their light curves vary on millisecond timescales. If one assumes that these photons are produced in a simple, optically thin region close to a compact object, two major problems arise. The first is the compactness problem: such a small and luminous source should trap its own radiation, as photons above 511 keV would readily collide and create electron–positron pairs ($\gamma + \gamma \rightarrow e^+ + e^-$). This would make the region opaque, preventing the escape of the high–energy gamma rays we in fact observe. The second is the spectral shape problem: the observed prompt spectra are broader than a simple blackbody and do not match the expectations of purely optically thin synchrotron emission either. Thus, neither a "thin" nor a "thick" model in isolation could explain the data.

To resolve this, Goodman (1986) and Paczyński (1986) independently proposed the relativistic fireball model. In this scenario, the central engine injects a vast amount of energy into a very small volume, creating a hot, dense plasma composed of photons and abundant electron–positron pairs. The system is initially optically thick: photons scatter so frequently on pairs that their mean free path is much smaller than the source size, trapping the radiation and thermalizing it into a blackbody distribution. Radiation pressure then drives the plasma to expand at ultra–relativistic speeds. As the fireball expands, the density falls and the optical depth decreases. Eventually, at a certain radius, the outflow becomes optically thin, meaning photons can escape without further scattering. This “last scattering surface” is called the photosphere, and the emergent radiation is expected to be quasi–thermal.

Studying photospheric models is therefore essential because they address the fundamental opacity problem in GRBs, provide a physical origin for thermal–like spectral components that are indeed observed in some bursts, and offer a natural baseline for more complex scenarios. While the pure fireball photosphere predicts a spectrum that is narrower than typically observed, it establishes a crucial foundation. Later refinements — such as multi–temperature photospheres, sub–photospheric dissipation, or hybrid models combining blackbody and synchrotron components — build directly on this picture. In short, photospheric fireball models are studied because they emerge as the natural consequence of the extreme physical conditions expected in GRBs, and they provide the theoretical starting point for connecting the central engine to the observed prompt emission (Goodman, 1986; Paczyński (pronunciation = pache-zinski), 1986).

Summary of Pe'er & Ryde (2011) – *A Theory of Multicolor Blackbody Emission from Relativistically Expanding Plasmas*

1. **The Problem:**
    - A pure blackbody (Planck) spectrum is **too narrow** to explain GRB observations.
    - Yet, we often see signs of a thermal peak in GRB data.
    - How can a thermal spectrum be broadened naturally, without adding "by hand" a second component?
2. **The Idea:**
    - In a **relativistically expanding jet** (a GRB fireball), photons do not all escape from the same radius or angle.
    - Photons coming from different parts of the outflow experience **different Doppler boosts**.
    - As a result, the observed spectrum is actually a **mixture (superposition) of many blackbodies** with slightly different temperatures.
    - This is called a **multi-color blackbody (mBB)**.
3. **How the Spectrum Changes:**
    - Instead of a narrow Planck shape, the combined emission is **broader**.
    - Especially at low photon energies, a single blackbody spectrum follows the Rayleigh–Jeans law, where the flux increases steeply with frequency (proportional to the square of the frequency). In contrast, the multicolor blackbody model predicts a much flatter behavior, where the flux remains nearly constant with frequency. This flatter low-energy slope is closer to what is actually observed in GRB spectra. This matches what we actually see in many GRB spectra.
4. **Time Evolution Predictions:**
    - After the central engine shuts off, the model predicts a characteristic cooling trend. The thermal flux decreases rapidly with time, roughly following an inverse–square decay, while the temperature cools more gradually, approximately following a power–law decline with time to the minus two–thirds.
    - These scalings agree with earlier observations by **Ryde (2004, 2005)** of thermal pulses in GRBs.

5. **Why it Matters:**
    - Shows that **you don't need exotic new physics** to broaden a thermal peak — geometry + relativistic motion are enough.
    - Provides a natural explanation for why thermal emission in GRBs looks broader than a single blackbody.
    - Bridges the gap between **theoretical fireball models** and **observed GRB spectra**.

In one line:

Pe'er & Ryde (2011) showed that the photosphere of a relativistically expanding GRB outflow produces a **multi-color blackbody spectrum** — broader and flatter than a single Planck function — simply due to angular and Doppler effects, naturally explaining the thermal-like components seen in GRBs.

Summary of Rees & Mészáros 2005; Pe'er et al. 2006

Dissipation Below the Photosphere and Its Effects on the Emerging Blackbody

The standard fireball model predicts that thermal radiation, advected from the central engine, should emerge at the photosphere with a nearly Planckian blackbody spectrum. However, in reality the outflow is rarely free of dissipation before reaching transparency. Both Rees & Mészáros (2005) and Pe'er, Mészáros & Rees (2006) emphasize that energy release occurring below or near the photosphere—through internal shocks, magnetic reconnection, or collisional heating (e.g., neutron–proton collisions)—profoundly modifies the emerging spectrum.

Role of Dissipation Mechanisms

- **Internal shocks**: Variability in the Lorentz factor of colliding shells injects non-thermal electrons below or near the photosphere.
- **Magnetic reconnection**: Converts Poynting flux into particle heating and turbulence, producing quasi-thermal electron populations.
- **Collisional heating**: For example, neutron–proton collisions deposit significant energy into the baryon–electron plasma.

In all cases, dissipation energizes electrons (and sometimes pairs), which then interact with the thermal photon field.

Comptonization of Thermal Photons

The dominant channel for energy transfer is inverse Compton (IC) scattering of the advected thermal photons. Even modest heating produces relativistic or semi-relativistic electrons, which upscatter the soft photons. As a result, the narrow Planck spectrum is transformed into a **Comptonized blackbody**: a thermal peak accompanied by a broadened high-energy tail. For moderate optical depths ($\tau \approx 0.1$–10), repeated scatterings produce a nearly flat $\nu F \nu$ spectrum above the thermal peak, extending up to the MeV range. This naturally resembles the observed Band-function high-energy slopes.

Optical Depth Dependence

- **Low τ (≈ 0.1–1):** The thermal peak survives near tens of keV, but IC scatterings broaden the spectrum upward, producing a flat or slightly rising $\nu F \nu$ tail up to about 10 MeV.
- **Intermediate τ (few – tens):** Multiple scatterings dominate, yielding a broad, flat $\nu F \nu$ component from ~30 keV to several hundred keV—similar to the observed prompt emission.

- **High τ ($\gtrsim$100):** Photons undergo many scatterings before escape. Non-thermal features are erased, and the spectrum relaxes to a Wien peak at 100 keV–1 MeV.

Pair Formation and Secondary Photospheres

Dissipation above or near the photosphere can generate photons with energies exceeding m_ec^2, leading to electron–positron pair cascades. Pairs increase the opacity and effectively shift the photosphere outward, creating a "pair photosphere." This boosts the luminosity of the emerging radiation and can soften the spectral peak into the X-ray or soft gamma-ray regime. Importantly, pair annihilation prevents runaway opacity: the number of pairs rarely exceeds the number of baryon-associated electrons by more than a factor of a few.

Spectral Features Produced

- A flattened high-energy slope ($\nu F_\nu \approx$ constant) above the thermal peak for $\tau \approx 0.1$–10.
- Steep low-energy slopes below the peak, which can naturally account for spectra harder than optically thin synchrotron predictions.
- Wien-like peaks at 100 keV–1 MeV at very high τ, consistent with observed clustering of GRB spectral peaks.
- High-energy cutoffs above ~100 MeV, arising from pair production and Klein–Nishina suppression.

Together, these explain why many GRBs show a quasi-thermal component plus a Band-like tail, without requiring pure optically thin synchrotron.

Key Physical Picture

The pristine blackbody advected from the central engine is never observed directly. Instead, sub-photospheric dissipation acts as a **spectral processor**:

- Mild heating → broadened thermal peak plus flat Comptonized tail.
- Intense heating at high τ → Wien peak.
- Pair creation → outward-shifted photosphere with modified peak location.

In summary, dissipation below the photosphere naturally converts a narrow Planck spectrum into a Comptonized blackbody with a thermal bump and a broadened high-energy extension. This framework accounts for the diverse prompt GRB spectra—flat νF_ν plateaus, steep low-energy slopes, Wien-like peaks, and high-energy cutoffs—with the precise outcome determined mainly by the optical depth and the efficiency of the dissipation process (Rees & Mészáros 2005; Pe'er et al. 2006).

Summary of Comptonized Blackbody (BB + PL/CPL)

Ryde (2004)
This study investigated the spectral evolution of individual GRB pulses using BATSE data. The analysis showed that many pulses could be fit with a blackbody component, sometimes with an additional power law. The blackbody temperature was observed to remain roughly constant at the start of a pulse and then decrease with time, following a decay close to a power law.

The author argued that this cooling pattern is consistent with thermal emission from the photosphere of the GRB outflow. The presence of an additional weaker non-thermal component in some cases suggested that GRB spectra are not purely thermal, but instead a combination of thermal and non-thermal processes. This work provided some of the first clear evidence that a photospheric blackbody can be directly identified in GRB prompt emission.

Ryde (2009)
The analysis of GRB 090902B using data from the Fermi GBM and LAT showed that the prompt spectrum was dominated by a strong, quasi-blackbody component. The emission was better described by a multicolor blackbody than by a single Planck function, with a characteristic temperature of about 290 keV. This thermal component contributed the majority of the observed flux during the early part of the burst.

From these fits, the study estimated a photospheric radius of approximately 1×10^{12} cm and a bulk Lorentz factor that varied but reached values up to around 750. The dominance of the thermal emission during the first seconds also explained the delayed onset of high-energy photons in the LAT compared with the GBM. This work established GRB 090902B as a key example of a burst where the photospheric component was the primary source of emission.

Guiriec (2011)
Fermi observations of GRB 100724B revealed that the Band function alone could not reproduce the spectrum accurately. By adding a blackbody with a temperature of around 45 keV, the fit improved significantly. The thermal component accounted for about 10% of the total flux and helped to correct anomalies in the Band parameters, such as shifting the peak energy to higher values.

The study emphasized that the thermal and non-thermal components showed independent variability, which strongly supported the idea of two physically distinct origins. The blackbody was interpreted as emission from the jet photosphere, while the Band component represented a separate non-thermal process. This was one of the first Fermi detections to clearly show a subdominant thermal component in a GRB spectrum.

Guiriec (2013)

The short GRB 120323A initially appeared unusual because a Band-only fit gave a peak energy of only about 70 keV, far lower than typical for short bursts. When a blackbody component with a temperature near 10 keV was included, the spectral fit improved and the Band peak energy shifted into the expected 250–300 keV range.

This adjustment also restored the well-known hardness–luminosity correlation that had seemed absent with the Band-only fit. The results showed that thermal emission can play an important role even in short GRBs, which had rarely shown such components before. This made GRB 120323A one of the first strong cases where photospheric emission was detected in a short burst.

Summary of Two-Component Jet (Spine–Sheath)

Lundman (2012)

This paper developed a theoretical framework for photospheric emission from relativistic, collimated outflows such as those in gamma-ray bursts. The authors investigated how the observed spectrum is shaped by jet geometry, especially when the Lorentz factor varies with angle. They found that photons emitted from different regions of the jet undergo different Doppler boosts, broadening what would otherwise appear as a simple blackbody. Instead, the result is a multicolor blackbody spectrum that can reproduce the non-thermal slopes often seen in GRB prompt emission.

A central focus of the study was the **spine–sheath jet structure**, in which a fast, narrow inner jet (spine) is surrounded by a slower sheath. The photospheric emission from such a structured jet naturally produces a two-component spectrum: a thermal peak arising from the spine's photosphere, and a broader, non-thermal-like component from the shear layer between spine and sheath. This interplay of components explains why many GRB spectra require both a blackbody and a Band-like fit, aligning with observations reported in several bursts.

The authors concluded that photospheric emission from structured, relativistic outflows can mimic the widely used empirical Band function without requiring additional dissipation processes. Their results demonstrate that a **combination of jet geometry and relativistic effects** is sufficient to produce the observed diversity of GRB prompt spectra. This work strongly supports the idea that thermal emission is a natural, inherent part of GRB prompt radiation and that the Band function may, in many cases, be a manifestation of broadened photospheric emission from spine–sheath jets.

Summary for Hybrid BB + Synchrotron

Burgess et al. (2014)

This paper presented a time-resolved spectral analysis of eight bright gamma-ray bursts detected by the Fermi Gamma-ray Burst Monitor. Unlike previous studies that relied mainly on empirical functions such as the Band model, the authors used physically motivated synchrotron radiation models and tested them against the data. They examined both slow-cooling and fast-cooling synchrotron scenarios to determine whether the observed prompt spectra could be consistently explained by electron acceleration and cooling in relativistic shocks.

The study found that in five of the eight GRBs, the synchrotron models alone could not account for the observed spectral shapes. In these cases, the addition of a blackbody component significantly improved the fits, with the blackbody contributing a measurable fraction of the total flux. The thermal component appeared as a distinct peak in the lower-energy portion of the spectrum, consistent with emission from the jet photosphere.

These results demonstrated that many GRB spectra are best described as a hybrid of a thermal blackbody and a non-thermal synchrotron component. The blackbody traced the physical conditions at the photosphere, while the synchrotron component captured the emission from accelerated electrons further out in the jet. By applying physical rather than empirical models, this work provided strong evidence that both photospheric and synchrotron processes act together during the prompt phase of many GRBs.

Yu et al. (2019)

This study introduced a Bayesian framework for time-resolved spectroscopy of GRB pulses, analyzing 38 individual pulses from 37 bursts observed by the Fermi/GBM. Using Bayesian blocks to define time bins, the authors obtained 577 spectra and fitted them with empirical models, mainly the cutoff power law (CPL) and the Band function. This was the first systematic application of a fully Bayesian approach to GRB spectroscopy, allowing for robust parameter estimation and model comparison.

The results showed that the CPL model was generally preferred over the Band function according to statistical criteria and posterior distributions. However, in several cases, empirical fits improved further when an additional blackbody component was included. This echoed earlier work showing that a purely non-thermal model is often insufficient to describe GRB spectra. In fact, the study confirmed that composite models combining a thermal component with a non-thermal function can more faithfully reproduce the observed diversity in GRB spectra.

An important finding was that about 60% of the analyzed spectra had low-energy slopes harder than the synchrotron “line of death,” making them incompatible with synchrotron emission alone. This strongly suggested the presence of thermal emission. The authors concluded that GRB prompt spectra often require a hybrid description: a blackbody representing photospheric emission, coupled with a non-thermal component such as synchrotron radiation. Their Bayesian catalog provided a new statistical foundation for investigating this thermal-plus-non-thermal picture of GRB emission.

Comprehensive Table of Observational and Theoretical Papers on GRB Photospheric Emission (Deep Research)

	Paper	Year	Type	Detailed Findings / Key Results
1	Preece et al.	1996	Observational	BATSE GRB spectral catalog (ApJS). Found ~15% of bursts had excess low-energy emission below 25 keV. Suggested possible soft thermal contribution, though not explicitly fit as blackbody. Early hint of thermal signatures.
2	Ryde	2004	Observational	Time-resolved BATSE analysis. Introduced BB+PL fits to pulses, with decaying blackbody temperature (t^(-2/3)). Showed thermal emission can be hidden in time-integrated spectra. Pioneering evidence for photospheric emission.
3	Ryde	2005	Observational	Reported several BATSE GRBs where Planck+PL fits improved spectra. Found kT ~30 keV components. Argued thermal emission may be a common feature of GRBs.
4	Abdo et al.	2009	Observational	Fermi discovery paper on GRB 090902B. Reported distinct extra spectral component plus high-energy LAT tail. Suggested a quasi-thermal peak of photospheric origin, though Band+PL fit was sufficient statistically.
5	Ryde et al.	2010	Observational	Detailed time-resolved analysis of GRB 090902B. Identified dominant multi-color blackbody (~290 keV) + PL. Derived photospheric radius ~1e12 cm, Lorentz factor ~750. Benchmark case for photospheric emission.
6	Guiriec et al.	2011	Observational	Fermi GRB 100724B required Band+BB fit. BB with kT ~45 keV contributed ~10% of prompt flux. Inclusion corrected anomalous Band indices. First Fermi-era robust detection of subdominant thermal component.
7	Ryde et al.	2011	Observational	Study of 6 GRBs (BATSE + Fermi) with strong thermal components. Classified 'Class I' bursts dominated by BB. Observed flux–temperature 7correlations consistent with dissipative photospheres.
8	Axelsson et al.	2012	Observational	GRB 110721A with extreme E_peak ~15 MeV. Early time bins required Band+BB, with kT ~1–2 MeV. BB contributed ~30% flux. Highest-temperature photospheric component reported in GRBs.

9	Zhang et al.	2012	Observational	GRB 120422A (low-luminosity GRB + SN). Swift-XRT early data showed thermal X-ray bump (kT ~1–2 keV). Interpreted as shock breakout from Wolf-Rayet star envelope rather than central photosphere.
10	Guiriec et al.	2013	Observational	Short GRB 120323A required Band+BB. BB with kT ~10 keV contributed 5–10% flux. Normalized short GRB parameters and revealed hardness–intensity correlation. First strong evidence of photospheric emission in short GRBs.
11	Ghirlanda et al.	2013	Observational	MNRAS study of Fermi bursts. Found BB evident throughout prompt emission of some GRBs. Derived Γ ~ few hundred, Rph ~1e12 cm. Supported idea that photospheric emission is ubiquitous in bright bursts.
12	Iyyani et al.	2013	Observational	GRB 110920A had narrow spectrum fit by multi-color BB + PL. BB dominated flux (>50%), with kT evolving 70→20 keV. Clear case of photosphere-dominated prompt emission.
13	Burgess et al.	2014	Observational	Survey of 37 Fermi GRBs with various models. Found 5 bursts with definitive BBs. Others consistent with synchrotron. Balanced view of thermal vs. non-thermal dominance.
14	Burgess & Ryde	2015	Observational/ Methodology	Investigated if blackbody detections are artifacts of spectral evolution. Showed integrated spectra can mimic BB, but time-resolved BBs are genuine. Cautionary methodological paper.
15	Nappo et al.	2017	Observational	Swift GRB 151027A had strong optical–X-ray flare at ~100s. Broadband fits required PL+BB (kT ~0.3 keV). BB contributed ~30% flux. Interpreted as photospheric flare or shock breakout.
16	Wang et al.	2023	Observational	GRB 220426A identified as thermal-radiation dominated. Multi-color BB fits spectra, nearly purely thermal. Derived Γ ~300–800, Rph ~1e11–1e12 cm. Modern clear case of photosphere-dominated GRB.
17	Chen & Peng	2024	Observational	GRB 220304A also thermal-dominated. Band+BB with BB ~50% of flux. Spectral sharpness evolution consistent with cooling photosphere. Confirmed existence of thermal GRB class.

18	Goodman	1986	Theoretical	Introduced cosmological fireball model. Predicted initial thermal flash from optically thick plasma. Laid foundation for photospheric emission in GRBs.
19	Thompson	1994	Theoretical	Proposed Alfvén turbulence heating below photosphere. Suggested multiple Compton scatterings broaden thermal spectrum. Precursor to dissipative photosphere models.
20	Mészáros & Rees	2000	Theoretical	Developed dissipative photosphere models. Showed moderate sub-photospheric heating broadens BB into Band-like spectrum. Explained X-ray rich GRBs as photosphere-dominated cases.
21	Rees & Mészáros	2005	Theoretical	Proposed delayed reheating below Rph as explanation for Band-like prompt emission. Offered solution to synchrotron 'line-of-death'. Key dissipative photosphere concept.
22	Pe'er et al.	2006	Theoretical	Detailed radiative transfer in dissipative outflows. Showed Comptonized photospheric emission reproduces Band spectra. Predicted quasi-thermal peaks + non-thermal high-energy tails.
23	Pe'er et al.	2007	Theoretical	Developed method to estimate Lorentz factor and initial radius from observed thermal components (kT, flux). Applied to BATSE bursts to derive Γ ~100–1000.
24	Beloborodov	2010	Theoretical	Collisional heating model: neutron-proton collisions below Rph generate pairs and photons. Produces photospheric component + GeV tail. Applied to GRB 090902B-like spectra.
25	Pe'er & Ryde	2010	Theoretical	Showed angle integration of BB emission yields non-thermal-like slopes (α ~ -1). Explained why photospheric spectra can look Band-like.
26	Lundman et al.	2013	Theoretical	Studied multi-color BB from jets with angle-dependent Γ. Showed photospheric emission can mimic variety of GRB spectra. Asked 'when does a blackbody look non-thermal?'.
27	Hascoët et al.	2013	Theoretical	Proposed that prompt GRB peaks can be explained purely by thermal emission in some cases. Extended models of photospheric emission to account for observed prompt spectra.

28	Beniamini & Giannios	2017	Theoretical	Examined polarization signatures of magnetized dissipative photospheres. Predicted weak (~few %) polarization for fireballs, higher for magnetized jets.
29	Beloborodov & Mészáros	2017	Review	Review summarizing state of photospheric emission models. Covered dissipative, collisional, and magnetized scenarios.
30	Pe'er & Ryde	2017	Review	Comprehensive review arguing thermal components may be nearly universal in GRBs. Summarized observational evidence and theoretical interpretations.
31	Burgess et al.	2020	Theoretical/Methodology	Introduced physical models (synchrotron + Comptonized thermal) fitted directly to GRB data. Advanced beyond empirical Band+BB fits to more physical frameworks.